Dear Professor Marelli,

Thank you for the opportunity to revise and resubmit our manuscript, *[CzeLex: A megastudy dataset of concreteness, imageability, age of acquisition, valence and gender ratings for 9,601 Czech words]* (Manuscript ID: [BR-Org-26-344.R1]). We are grateful to you and the reviewers for the positive and constructive feedback, which has been invaluable in strengthening our work.

We have carefully addressed all of the comments provided during this second round of reviews. In doing so, we have revised the manuscript to correct the minor comments provided. Our point-by-point response to each reviewer's comment is included below. Following your guidance, all edits to the manuscript are presented in blue font, as are the responses to the reviewers.

We believe these revisions have improved the quality of the manuscript, and we hope that it will now be suitable for publication in Behavior Research Methods.

Thank you again for your time and consideration.

Sincerely,

James Brand (on behalf of Filip Smolík, Jan Chromý, Markéta Cehánková and Mikuláš Preininger)

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Reviewer: 1

Comments to the Author

The authors have carefully addressed my comments. The manuscript has been improved substantially, particularly regarding the introduction, methodological clarity, and validation analyses. I remain somewhat cautious about the use of the random intercepts as lexical estimates, and I would have appreciated a slightly more detailed discussion of the conceptual issues surrounding the BLUP/random-effect estimation in the manuscript itself. The finding that the results are nearly identical to analyses based on arithmetic means is encouraging, but it does not serve as theoretical explanation. If the conclusions are essentially identical under both approaches, the manuscript should more clearly justify why the model-based estimates are preferable. Otherwise, readers may reasonably wonder whether the additional methodological complexity is justified for the present purpose. However, I do leave it to the authors to keep the method of their choice. Nevertheless, I strongly recommend that the authors devote more attention (couple of sentences, not more) in the manuscript to the methodological debate surrounding the interpretation of BLUPs/random-effect estimates, not just refer to a new paper which is in preparation.

Finally, and crucially, the authors should rewrite the paragraph they introduced on the topic, as I believe that the use of the cited papers is not fully adequate. For example, the claim that „there are mixed views on the suitableness of using both the random intercepts (see Houslay & Wilson, 2017; Kliegl et al., 2011) and the mean (Baayen et al., 2008; Drager & Hay, 2012)“ is misleading. Both Houslay & Wilson, 2017 and Kliegl et al., 2011 discuss limitations and potential misuse of BLUPs/random effects rather than providing evidence for the existence of mixed views on the suitability of random intercepts. Likewise, Baayen et al., 2008, and Drager & Hay, 2012 advocate the use of mixed-effect models for statistical inference, but, to the best of my knowledge, do not argue that arithmetic means are inappropriate descriptive summaries. Therefore, these citations do not support the claim being made. Therefore, I consider the manuscript acceptable for publication, but I recommend that, prior to publication, the authors revise this paragraph because it overstates what the cited methodological papers actually say.

We thank the reviewer for their comments and helpful input on this point. We have added the suggested discussion on the methodological debate and hope that this added detail will improve clarity on the subject. We have also addressed the use of the citations to more accurately reflect the arguments being made. We have also added details on the justification for choosing the intercepts based models too, alongside notes of caution for other researchers. We hope this suitably addresses the reviewers concerns and provides enough detail for readers to gain a better understanding on the topic.

The text on p.23 has been updated to:

“For reference, as we were using measures for RTs and accuracy derived from mixed-effects modelling, instead of the traditional descriptive statistic of the mean, we conducted parallel analyses of all the results reported below and found no indication that the two approaches led to any differences in the pattern of results, indeed they were almost identical in terms of the general patterns being reported. This lack of a difference between the results may be explained by the fact that our analysis had very large samples at the participant and item level, so variance within the intercepts and traditional mean

measures would consequently both be low, compared to an analysis with smaller samples. Nevertheless, it is important to highlight that the use of extracting random intercepts for a secondary analysis, and the added complexity involved when using them in this way, needs to be justified.

In our analysis, we focussed on the intercepts derived estimates as our sample sizes were sufficiently large. Additionally, our semantic norms were also derived from intercepts derived estimates, where participant-level variation was being considered in the models, so this provided us with a unified approach across our estimates. The practice of extracting random intercepts for analyses has been suggested to be a useful approach when considering alternative ways of obtaining item-level estimates for analyses (see e.g. Drager & Hay, 2012), however there are also reasons to be cautious about the approach. For example, Houslay and Wilson (2017) and Kliegl et al. (2011) both discuss the limitations of using this approach as the estimates do not carry forward the error from the estimates themselves, which can lead to an increased likelihood of obtaining false positives in subsequent analyses that use the intercepts. Whilst the results from both the intercepts and traditional mean do not appear to represent any substantial divergence, future work may investigate the conditions under which different methods of obtaining estimates may, or may not, lead to different outcomes in the analyses."

Reviewer: 2

Comments to the Author

I thank the authors for addressing the comments and revising the manuscript where needed. I have spotted just two minor points:

Thanks for your input on the manuscript during the review process, very much appreciated.

- A typo in a section “Dimensions being normed” with double full stops (p.7, l. 14). It might be worth splitting this long sentence into two.

Done.

- Why does the total number of participants stated in the Abstract (N=4391) not correspond to the number reported in the “Participants section” (p. 10) and Figure 2? I apologize if I have missed something here.

Thanks for flagging this, we have updated the abstract to 4,405 which was the final sample size.

Apart from this, I have no further comments and believe that this megastudy will be a useful addition to the existing sources in the psycholinguistic literature.